

Reading & References

A compendium for the Field Trilogy

Anima · Numen · Limen · Fragile Light

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1. Scholarly articles & papers — Hard problem, free will, consciousness theory

The theoretical core. Peer-reviewed and pre-print research the trilogy's framework rests on.

Hard Problem and Free Will: an information-theoretical approach ↗

Giacomo Mauro D'Ariano & Federico Faggin (arXiv, 2020)

The foundational paper for the trilogy's *knowing, choosing, feeling* framework. D'Ariano (physics) and Faggin (the engineer who invented the microprocessor) argue that consciousness and free will are not emergent from matter but irreducible properties of quantum information itself. The information-theoretic backbone of the trilogy's argument.

Universal consciousness as foundational field: A theoretical bridge between quantum physics and non-dual philosophy ↗

Maria Strømme, AIP Advances (2025)

The first peer-reviewed mathematical formalization of consciousness as a fundamental Φ -field, with individual minds as localized excitations of that field — analogous to photons as excitations of the electromagnetic field. *Limen's* mystics chapter rests on this paper.

"Facing Up to the Problem of Consciousness" ↗

David J. Chalmers, Journal of Consciousness Studies 2(3) (1995)

The paper that named the hard problem. Chalmers's 1995 essay drew the line that has organized the field ever since: the *easy* problems (information processing, integration, reportability) are tractable in principle; the *hard* problem — why any of this should be accompanied by experience at all — is different in kind, and no purely physical description of the brain can derive it. Every consciousness framework, including the trilogy's, has to answer it. The PDF at the link is the canonical full-text version, hosted on Chalmers's own site.

"Towards a neurobiological theory of consciousness" ↗

Francis Crick & Christof Koch, Seminars in the Neurosciences 2 (1990)

The founding paper of the modern *neural correlates of consciousness* (NCC) program. Crick (DNA) and Koch (neuroscience) propose that consciousness can be made tractable for systems neuroscience by narrowing the question to visual awareness and looking for the cortical activity patterns — recursive, re-entrant, sustained — that accompany what enters perception. The orthodox empirical view the trilogy takes seriously and reads differently: the same correlations the productionist sees as evidence of generation, the receiver model reads as evidence of *reception*.

"Are You Living in a Computer Simulation?" ↗

Nick Bostrom, Philosophical Quarterly 53(211) (2003)

The simulation trilemma. Bostrom's formal probabilistic argument: at least one of three propositions is almost certainly true — civilizations almost always go extinct before reaching the technological maturity to run ancestor-simulations, civilizations that *can* run them almost always choose not to, or we are very probably already living inside one. The argument that turned the simulation hypothesis from science fiction into a respectable item in contemporary metaphysics, and the argumentative ancestor of every nested-reality scene in *Numen* — the question *Limen* refuses to dismiss. PDF hosted at Bostrom's simulation-argument.com.

"Taking AI Welfare Seriously" ↗

Robert Long, Jeff Sebo, Patrick Butlin et al. (incl. David Chalmers), *arXiv:2411.00986* (2024)

Chalmers and colleagues arguing that the moral status of AI systems can no longer be deferred. The published companion to *Numen's* hybrid arc.

2. Scholarly articles & papers — Quantum foundations & nonlocality

"On the Einstein Podolsky Rosen paradox" ↗

John S. Bell, *Physics* 1(3) (1964)

Bell's inequality. The six-page paper that turned the EPR philosophical objection into something a laboratory could test. Bell formalized Einstein, Podolsky and Rosen's 1935 argument that quantum mechanics needed "hidden variables" to be complete — and proved that no local hidden-variable theory could reproduce quantum mechanics' statistical predictions. The mathematical demonstration is exact: any local realist account must satisfy an inequality which quantum mechanics violates for appropriately chosen measurement angles. Half a century of experiments — Clauser, Aspect, Zeilinger — have ruled in favor of quantum mechanics, closing every major loophole. The paper that ended the Einstein-Bohr debate and opened the empirical program that became the 2022 Nobel. Hosted open-access at the APS archive of *Physics Physique Физика*.

"Experimental Realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A New Violation of Bell's Inequalities" ↗

Alain Aspect, Philippe Grangier & Gérard Roger, *Physical Review Letters* 49(2) (1982)

The first decisive experimental closure of Bell's inequalities in an EPR-Bohm-style optical setup with two-channel polarizers — nonlocality as empirical fact, not interpretation. The companion paper later that year, with Dalibard and Roger, added time-varying analyzers to tighten the locality case. Together the two 1982 papers crossed the threshold of reasonable doubt and underwrote the 2022 Nobel Prize. The result behind every entanglement scene in the trilogy, and the empirical floor under *Limen's* non-separable field cosmology. PDF hosted open-access at the APS *Physical Review Letters* archive.

Scientific Background on the Nobel Prize in Physics 2022 — "For experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science" ↗

The Nobel Committee for Physics, Royal Swedish Academy of Sciences (2022)

The official scientific background to the prize awarded to Clauser, Aspect, and Zeilinger. The clearest single document on what Bell's theorem actually rules out, how the loopholes were closed, and what the experiments leave standing.

Spooky Action at a Distance: The Phenomenon That Reimagines Space and Time ↗

George Musser, Farrar, Straus and Giroux (2015)

A senior editor at *Scientific American* walks through Bell, EPR, the holographic principle, ER=EPR, and the loop-quantum-gravity programs to make a single argument: that the macroscopic "expanse where events happen in absolute locations" is an appearance, not a deep feature of the world. As the publisher's own description puts it — "*multiple branches of physics now suggest that, at a deeper level, there may be no such thing as place and no such thing as distance.*" The clearest book-length statement of the claim that distance is rendered, not fundamental.

Entanglement and the Geometry of Spacetime ↗

Juan Maldacena, Institute for Advanced Study (2013)

The accessible essay version of Maldacena and Susskind's ER=EPR conjecture: every entangled pair is connected by a microscopic Einstein-Rosen bridge — a wormhole — and spacetime itself is *woven* by entanglement. If this is right, two particles a billion light-years apart are not actually far apart; they are geometrically connected by a non-traversable wormhole whose existence *is* the entanglement. Distance, in that picture, is the macroscopic appearance of a deeper connectivity graph. The technical anchor under *Limen's* argument that locality is a rendering.

Minimal Length Scale Scenarios for Quantum Gravity ↗

Sabine Hossenfelder, *Living Reviews in Relativity* 16, 2 (2013)

The definitive literature review on whether the Planck length is a fundamental minimum. Hossenfelder walks through every independent line of argument — the generalized uncertainty principle, deformed special relativity, loop quantum gravity, string theory's T-duality, the black-hole thought experiments — and shows that they all converge on the Planck scale as a lower bound on physically meaningful distance. The convergence of independent approaches is itself the strongest piece of evidence.

Discreteness of area and volume in quantum gravity ↗

Carlo Rovelli & Lee Smolin, *Nuclear Physics B* 442, 593 (1995)

The technical anchor for the claim that *geometry itself comes in quanta*. Rovelli and Smolin proved that the area and volume operators in loop quantum gravity have *discrete spectra*, with the smallest non-zero area being a numerical multiple of the Planck area (ℓ_p^2). Not "we cannot measure below the Planck scale" — but "there are no states below it." Space is granular at 10^{-35} m. The technical scaffold beneath *Limen's* argument that the universe is rendered at a finite resolution.

"Universal upper bound on the entropy-to-energy ratio for bounded systems" ↗

Jacob D. Bekenstein, *Physical Review D* 23, 287 (1981)

A separate route to the same conclusion: the information content of any finite region of space has a hard upper bound, proportional to size $\times$ energy. Counting bits at that bound forces a Planck-area pixel structure on bounding surfaces — the foundation of what later became the *holographic principle* ('t Hooft, Susskind, Maldacena). Reality has a finite information density, set by the Planck scale, and black holes saturate the bound — they are the ultimate information-storage devices nature allows.

Reality Is Not What It Seems: The Journey to Quantum Gravity ↗

Carlo Rovelli, Riverhead Books (2017)

The accessible companion to the Rovelli-Smolín technical paper above. One of the founders of loop quantum gravity walks a general reader through what it means to say "*space is made of quanta*" — the spin networks, the Planck-scale granularity, the absence of a continuous background. Chapter 7 in particular is the cleanest popular explanation of why the Planck length and Planck time are not just convenient units but appear to be a *floor* on physical reality.

"The Two-State Vector Formalism: An Updated Review" ↗

Yakir Aharonov & Lev Vaidman, *Lecture Notes in Physics* 734 (2008)

The time-symmetric reformulation of quantum mechanics — at any intermediate moment the system is described by *two* state vectors, one evolving forward from preparation and one evolving backward from the final measurement. Pre- and post-selected systems display anomalous "weak values," apparent particles-in-two-boxes, and other patterns that look paradoxical under standard forward-only QM but become natural when both boundary conditions matter. The technical scaffold beneath *Limen*'s argument that the universe is read both ways, and the formal language for the trilogy's symmetric 300-ms gaps (readiness potential on one side, Lucía Reyes's cymatic pre-event window on the other).

"The Amplituhedron" ↗

Nima Arkani-Hamed & Jaroslav Trnka, *Journal of High Energy Physics* (2014); *arXiv:1312.2007*

Planar scattering amplitudes in $N=4$ super Yang-Mills computed as the canonical differential form on a positive geometric object — the amplitude is literally the geometry's volume form. Spacetime locality and unitarity, the most foundational-looking principles of physics, emerge from the combinatorial structure of the geometry's boundaries rather than being assumed as primitive axioms. The cleanest existing demonstration of the slogan *spacetime is doomed* — not in the sense that there is no spacetime, but in the sense that spacetime is not the ground floor. The geometry-precedes-physics argument *Limen* leans on.

"Experimental realization of Wheeler's delayed-choice gedanken experiment" ↗

Vincent Jacques et al., *Science* 315(5814) (2007)

Single-photon Mach-Zehnder interferometer with a quantum-random-number generator inserting or removing the output beamsplitter *after* the photon has already entered, at space-like separation. The closed configuration gives ~94% interference visibility (wave behavior); the open configuration gives which-path information with <1% error (particle behavior). Each photon's detection statistics match the configuration chosen after it entered — the future genuinely participates in the constitution of the past. The empirical anchor for *Limen*'s observer chapters, and the cleanest experimental window onto the time-symmetric quantum mechanics formalized by Aharonov and Vaidman.

"Wheeler's delayed-choice gedanken experiment with a single atom" ↗

A. G. Manning, R. I. Khakimov, R. G. Dall & A. G. Truscott, *Nature Physics* 11, 539–542 (2015)

The single-atom version of the delayed-choice result. Manning and colleagues realized Wheeler's experiment with a single metastable helium atom in a matter-wave Mach-Zehnder interferometer, with the open/closed configuration chosen by a random number generator triggered *after* the atom had already passed the first beam splitter. Closed: high-visibility interference (wave). Open: which-path information (particle). The configuration chosen after entry determines which behavior appears — for a massive particle, not just for light. The cleanest experimental form of the question *Limen* asks about time.

3. Scholarly articles & papers — Anomalous evidence

***Terminal lucidity: a review and a case collection* ↗**

Michael Nahm, Bruce Greyson, Emily Williams Kelly, Erlendur Haraldsson · *Archives of Gerontology and Geriatrics* (2012)

A collection of historical case reports documenting unexpected returns of coherence in patients with profound cortical compromise, drawn from medical literature dating back to the 19th century. Read about the research →

"Neural filters to conscious awareness and the phenomena that reduce their impact" ↗

Marjorie Woollacott & Marina Weiler

A neuroscience review of the gating mechanisms by which the brain limits what reaches conscious awareness — directly relevant to the trilogy's antenna/filter model of perception.

"Restructuring consciousness - the psychedelic state in light of integrated information theory" ↗

Carhart-Harris et al., *Frontiers in Human Neuroscience* (2015)

An IIT-framed analysis of psychedelic states as a relaxation of the brain's filtering architecture — integrated-information content rising as model-constraints fall.

"The savant syndrome: an extraordinary condition. A synopsis: past, present, future" ↗

Darold A. Treffert, *Phil. Trans. R. Soc. B* 364(1522), 1351–1357 (2009)

The canonical short review of savant phenomenology. Capacities that arrive without learning, supported by phenomenal domain-specific memory, sometimes congenital (autism plus an island of genius), sometimes *acquired* after brain injury, stroke, or frontotemporal dementia — capacity appearing where there was none before, in a previously neurotypical person. Treffert's framing: no model of brain function is complete unless it can absorb the jarring juxtaposition of severe disability with spectacular ability. Direct evidence that the antenna sometimes receives more than the producer-model brain can explain.

"Surfaces and interfacial water: Evidence that hydrophilic surfaces have long-range impact" ↗

Jian-Ming Zheng, Wei-Chun Chin, Eugene Khijniak, Eugene Khijniak Jr. & Gerald H. Pollack, *Advances in Colloid and Interface Science* 127(1) (2006)

The empirical paper behind Pollack's fourth-phase water claim — long-range structuring at hydrophilic interfaces. Solute-free "exclusion zones" extending *hundreds of microns* from the surface (six orders of magnitude beyond the classical few-nanometer expectation), characterized by altered mobility, hydrogen-bonding, electrical potential, and IR/UV/NMR signatures. The biology where field effects become physically measurable — and a candidate substrate for QED coherence domains, microtubule quantum effects, and the macroscopic structured-water field on which *Limen's* antenna model rests. Read the consolidated explainer — EZ water, cellular memory, and microtubule coherence →

"Effect of Health-Promoting Agents on Exclusion-Zone Size" ↗

Gerald H. Pollack et al., *Dose-Response* 16(3) (2018)

Nutraceuticals (turmeric, holy basil, probiotics, coconut water) and common analgesics (aspirin, acetaminophen) expand EZ water at physiological-ish doses over wide hormetic ranges; glyphosate (Roundup) shrinks it monotonically across all tested concentrations, falling to $\sim\frac{1}{5}$ of control near regulatory limits. The dose-response paper behind the EZ-water phenomenology — what enhances life also expands the structured water around it, what harms life contracts it. A candidate mesoscale physical correlate of biological effect, and the experimental hinge between EZ physics and the trilogy's receiver model. Read the consolidated explainer →

"EZ Water and the Origin of Life" ↗

EZ-water research program (2022)

The boldest frontier extension of the EZ-water program: the proposal that ordered, charge-separated, liquid-crystalline EZ water in pre-cellular aggregates might have served as an information-bearing scaffold *preceding* genetic polymers — a pre-RNA form of memory. Highly speculative, not established in mainstream cell biology, and worth knowing about precisely because it marks how far out the speculation now extends. The receiver model the trilogy defends is more conservative, but it shares the same intuition: the body is a coherent, field-sensitive medium, and the water in it is not inert. Read the consolidated explainer →

4. Scholarly articles & papers — Brain, attention, meditation, bioelectricity

***The Computational Boundary of a "Self": Developmental Bioelectricity Drives Multicellularity and Scale-Free Cognition* ↗**

Michael Levin, *Frontiers in Psychology* (2019)

The foundational theoretical paper of Levin's program at Tufts. Cognition — memory, goal-directedness, problem-solving — is not unique to brains; it is what bioelectric networks do at every scale. The brain is a specialization of an older and broader phenomenon. The trilogy's "body as antenna" image is rebuilt here in cellular terms.

Bioelectric networks: the cognitive glue enabling evolutionary scaling from physiology to mind ↗

Michael Levin, *Animal Cognition* (2023)

The extension of the framework explicitly to cognition. Bioelectric signaling was doing cognitive work in cells and tissues for billions of years before neurons existed; brains are a specialization of pre-existing bioelectric intelligence, not its origin. The most direct evidence yet that the receiver model is structural at the cellular scale.

The Multiscale Wisdom of the Body: Collective Intelligence as a Tractable Interface for Next-Generation Biomedicine ↗

Michael Levin, *BioEssays* (2025)

The clinical-facing version. If the body is a self-organizing problem-solving system, the role of medicine shifts: from repairing broken parts to communicating with the body's existing intelligence and providing the conditions it needs to repair itself. The medicine *Anima's* narrator is, by the end of the novel, learning to practice.

"Time of conscious intention to act in relation to onset of cerebral activity (readiness-potential). The unconscious initiation of a freely voluntary act." ↗

Benjamin Libet, Curtis A. Gleason, Elwood W. Wright & Dennis K. Pearl, *Brain* 106(3) (1983)

The classic readiness-potential experiment. EEG shows that the brain begins preparing a "freely voluntary" finger movement about 350 ms before the subject reports being consciously aware of deciding to move. Used for decades as the canonical empirical brief against free will — and the source of one of the trilogy's central reframings: the 300-ms gap is not the absence of agency but the latency of the field rendering local choice through biological tissue.

"An accumulator model for spontaneous neural activity prior to self-initiated movement" ↗

Aaron Schurger, Jacobo D. Sitt & Stanislas Dehaene, *PNAS* 109(42) (2012)

The most-cited methodological reinterpretation of Libet. Schurger and colleagues model the readiness potential as the average of stochastic neural fluctuations integrated by a leaky accumulator until they cross threshold — meaning the RP is the statistical signature of noise being shaped by averaging, not a deterministic unconscious "command" beginning 350 ms before awareness. Quietly reopens a question the orthodox Libet reading had appeared to close.

"Recovery from the Passage of an Iron Bar through the Head" ↗

John M. Harlow, *Publications of the Massachusetts Medical Society* 2 (1868)

Harlow's original case report on Phineas Gage. The founding document of the neurology of personality.

"Loss of recent memory after bilateral hippocampal lesions" ↗

William B. Scoville & Brenda Milner, *J. Neurology, Neurosurgery & Psychiatry* 20(11) (1957)

The H.M. paper. The clinical event that founded modern memory research.

"Some Effects of Disconnecting the Cerebral Hemispheres" (Nobel Lecture) ↗

Roger Sperry (1981)

Sperry's Nobel address on split-brain patients — the original empirical basis for McGilchrist's hemispheric thesis.

"The Return of Phineas Gage: Clues about the brain from the skull of a famous patient" ↗

Hanna Damasio et al., *Science* 264(5162) (1994)

The 1994 reconstruction of Gage's injury — frontal-lobe damage as a window onto personality and choice.

"The free-energy principle: a unified brain theory?" ↗

Karl Friston, *Nature Reviews Neuroscience* 11(2) (2010)

Friston's unified account of perception, action, and learning as variational free-energy minimization. The most mathematically serious version of the predictive-processing brain.

"Meditation experience is associated with differences in default mode network activity and connectivity" ↗

Judson A. Brewer et al., *PNAS* 108(50) (2011)

The neural signature of long-term meditation — quieting of the default-mode network. The brain learning to stop generating itself.

"Long-term meditators self-induce high-amplitude gamma synchrony during mental practice" ↗

Antoine Lutz, Lawrence L. Greischar, Nancy B. Rawlings, Matthieu Ricard & Richard J. Davidson, *PNAS* 101(46) (2004)

The Tibetan monk gamma study — sustained 40 Hz synchrony at amplitudes never recorded outside meditation. Empirical evidence that attention is a trainable instrument.

"Attention regulation and monitoring in meditation" ↗

Antoine Lutz, Heleen A. Slagter, John D. Dunne & Richard J. Davidson, *Trends in Cognitive Sciences* 12(4) (2008)

The taxonomy of attention practices — focused-attention vs. open-monitoring — and what each does to the brain. The contemplative-science scaffold of *Limen's* mystics chapter.

"Leisure activities and the risk of dementia in the elderly" ↗

Joe Verghese et al., *NEJM* 348(25) (2003)

Cognitive engagement as protection against cognitive decline — the clinical evidence that what the antenna does, it stays able to do.

"Teresa of Ávila and the Phenomenology of Mystical Experience"

J. López-Gay, *Revista de Espiritualidad* 74 (2015)

The longer-form companion to Marín García's *Words for the Gift*: López-Gay reads the seven mansions of the *Castillo Interior* as an ordered state-space of altered consciousness, with a typology of paradigmatic experiences (prayer of quiet, rapture, intellectual vs. imaginative visions, vision of hell) and a discernment grammar for telling them apart. The six structural moves are: bracketing metaphysics, treating the texts as first-person data, the mansions as ordered phenomenology, intentionality without sensory imagery, embodiment and affectivity, and Teresa's own discernment-by-fruits as proto-phenomenological rigor. The framework that lets the trilogy juxtapose Teresa with Advaita's *sahaja samādhi* and the receiver model with contemporary self-layering theory. Read the integrated explainer — López-Gay's seven-mansion architecture and Marín García's gift structure, together →

"Words for the Gift: A Phenomenological Account of St. Teresa of Ávila's Religious Experience"

Sergio Marín García

The cleanest contemporary phenomenological reading of Teresa's *Libro de la Vida*. Marín García argues that Teresa's distinction between *acquired* and *infused* prayer is not a difference of intensity but a difference of *mode of givenness*: the first is intentional in Husserl's strict sense (the soul aims at God-as-object), the second is what Marion calls *counter-intentional* (the soul receives what it did not produce). The five marks of the infused mode — surprise, novelty, suddenness, absoluteness, inter-personal origin — are exactly the structural marks the trilogy ascribes to receiver-mode events. Teresa's warning against *noise* — "running about with the intellect looking for many words and reflections" — is the sharpest description in the literature of what a productive brain does to a receiver.

5. Scholarly articles & papers — Frequency, music, and the body

"Hearing. II. The physical basis of the action of the cochlea" ↗

Thomas Gold, *Proc. Royal Society B* 135 (1948)

Gold's prediction of active amplification inside the cochlea — forty years before it was confirmed. The cochlea as instrument, not microphone.

"The influence of cochlear shape on low-frequency hearing" ↗

D. Manoussaki et al., *Physical Review Letters* 96 (2006)

The physics demonstration that the cochlea's spiral isn't decorative. Why ears are shaped the way they're shaped.

"Cochlea's graded curvature effect on low frequency hearing" ↗

D. Manoussaki et al., *PNAS* 105(16) (2008)

The mathematical demonstration that the cochlea's curvature itself does work at low frequencies. Geometry as biology.

"Auditory beats in the brain" ↗

Gerald Oster, *Scientific American* 229(4) (1973)

The classic 1973 introduction of binaural beats. Two pure tones of slightly different frequency presented one to each ear — the brain perceives a third tone at the difference frequency. The third tone is not physically present in the air; it is generated centrally, almost certainly at the superior olivary complex in the brainstem where inputs from the two ears first converge. Oster establishes the perceptual window (carrier < ~1000 Hz, difference < ~30 Hz), frames binaural beats as a probe of how the brain computes interaural phase for sound localization, and proposes their use as a diagnostic for central auditory function (with reported sensitivity to Parkinson's, hormonal state, and other systemic conditions). Resonance as fact, not metaphor — the brain itself is the resonator.

"Singing modulates mood, stress, cortisol, cytokine and neuropeptide activity in cancer patients and carers" ↗

Daisy Fancourt et al., *ecancermedicalscience* 10:631 (2016)

Empirical measurement of what every choir already knows. A larger, clinically meaningful population — cancer patients and their caregivers — showing that a single group-singing session reduces cortisol and inflammatory cytokines and raises mood-linked neuropeptides. The dose-response data behind the trilogy's chord chapters. Read the consolidated singing explainer →

"Changes in mood, oxytocin, and cortisol following group and individual singing: A pilot study" ↗

Arla Good & Frank A. Russo, *Psychology of Music* 50(4) (2022)

The cleanest pilot dissection of *which part* of singing carries the affective payload. Within-subjects design: the same participants did both group and solo singing. **Cortisol** dropped in both conditions — singing itself is stress-regulating. **Oxytocin** rose only after group singing. **Mood** improved only after group singing. Critically, the changes in *oxytocin* — not cortisol — tracked the changes in mood. This implicates oxytocin-linked social bonding, not vocal-motor activity alone, as the mechanism of the "singing high." The endocrine signature of *resonance with another*, not just resonance. Read the consolidated singing explainer — the four studies together →

"Does singing promote well-being?: An empirical study of professional and amateur singers during a singing lesson" ↗

Christina Grape et al., *Integrative Physiological & Behavioral Science* 38(1) (2003)

The early empirical pilot, comparing 8 amateurs and 8 professionals across a single singing lesson with ECG, blood draws, visual-analogue scales, and a semi-structured interview. **Oxytocin rose in both groups**; cortisol and prolactin showed sex-specific responses (up in men, down in women); heart-rate-variability power rose during singing in professionals but not amateurs (the "heart-brain connection" of trained singers). On the affective side, amateurs reported increased joy and elation that professionals did not; both groups felt more energetic and relaxed. The interview material revealed that professionals approached lessons achievement-oriented (technique, vocal apparatus) while amateurs framed them as self-expression and emotional release — a frame difference that may explain the affective dissociation. Read the consolidated singing explainer — the four studies together →

"Effects of choir singing or listening on secretory immunoglobulin A, cortisol, and emotional state" ↗

Gunter Kreutz et al., *Journal of Behavioral Medicine* 27(6) (2004)

The immune-system arm of the singing literature. Choir members who sang showed elevated secretory IgA (a mucosal-immunity marker) and reduced cortisol; those who only *listened* to the same music showed neither effect. The action is in the producing, not the receiving. Read the consolidated singing explainer →

"Psychobiological Effects of Choral Singing on Affective State, Social Connectedness, and Stress: Influences of Singing Activity and Time Course" ↗

Antje Bullack, Carolin Gass, Urs M. Nater & Gunter Kreutz, *Frontiers in Behavioral Neuroscience* 12:223 (2018)

The Kreutz group's longer-form replication and extension of the 2004 paper. Mixed amateur choir, two sessions one week apart (singing vs. listening, same repertoire), saliva and affect measures pre- and post-. Singing increased positive affect, reduced negative affect, and raised secretory IgA. Listening reduced cortisol but *paradoxically increased negative affect*. The dissociation between endocrine and subjective measures is the methodologically important result: stress-hormone reduction and mood improvement are separable. Active engagement matters in a way passive exposure does not. Read the consolidated singing explainer →

"Effects of Music Training on the Child's Brain and Cognitive Development" ↗

Gottfried Schlaug, Andrea Norton, Katie Overy & Ellen Winner, *Annals of the NY Academy of Sciences* 1060 (2005)

The longitudinal-plasticity arm of the music literature. Schlaug and colleagues review a decade of adult-musician neuroimaging (enlarged corpus callosum in early-trained musicians, increased gray matter in motor and auditory cortices, more focused fMRI activations during melodic and rhythmic tasks) and add their own preliminary longitudinal data on children: after just 14 months of instrumental lessons, 5–7-year-olds show gray-matter increases in motor and auditory regions plus measurable improvements in fine motor control and melodic discrimination. Cross-sectional data on 9–11-year-olds with several years of lessons add modest far-transfer effects in vocabulary and motor coordination. The plasticity argument behind *Limen's* claim that the antenna is tunable — and the developmental complement to the acute-endocrine singing findings. Read the consolidated music-and-body explainer →

"The physiological principle of minimum work" ↗

Cecil D. Murray, *PNAS* 12(3) (1926)

Murray's law — the geometric optimum of branching vasculature. Biology shapes itself by minimizing what it spends to transmit.

6. Scholarly articles & papers — Bioelectromagnetism, neurofeedback, the body in signal

"Augmentation of bone repair by inductively coupled electromagnetic fields" ↗

C. Andrew L. Bassett et al., *Science* 184 (1974)

The clinical demonstration that bone healing responds to electromagnetic fields — biology answering to invisible structure.

"Anti-inflammatory properties of the vagus nerve" ↗

Bruno Bonaz, Valérie Sinniger & Sonia Pellissier, *Journal of Physiology* 594(20) (2016)

Vagal tone as anti-inflammatory in measurable terms. The clinical pathway by which the body's listening modulates its inflammation.

"The Energetic Heart: Bioelectromagnetic Interactions Within and Between People" ↗

Rollin McCraty & Doc Childre, in *Clinical Applications of Bioelectromagnetic Medicine* (Rosch & Markov, eds., 2003); hosted at the HeartMath Institute

The HeartMath measurements — cardiac electromagnetic fields detectable between people in proximity. The biology where the field hypothesis becomes physically testable.

"Gamma frequency entrainment attenuates amyloid load and modifies microglia" ↗

Hannah F. Iaccarino et al., *Nature* 540 (2016)

40 Hz light stimulation clearing amyloid plaque in mouse models — frequency as therapy at the cellular level. The Tsai-lab paper behind *Limen's* gamma chapter.

"Multi-sensory gamma stimulation ameliorates Alzheimer's-associated pathology and improves cognition" ↗

Anthony Martorell et al., *Cell* 177(2) (2019)

The follow-up combining light and sound at 40 Hz — broader pathology rescue, including in tau models. The translational arm of the gamma-entrainment program.

"Efficacy of neurofeedback treatment in ADHD" ↗

Martijn Arns et al., *Clinical EEG and Neuroscience* 40(3) (2009)

The meta-analytic case for neurofeedback in ADHD. The clinical evidence that attention is mechanically trainable.

"What is neurofeedback: An update" ↗

D. Corydon Hammond, *Journal of Neurotherapy* 15(4) (2011)

The review article for clinicians — how neurofeedback works, what it does, where the evidence stands.

"Suppression of seizures in an epileptic following sensorimotor EEG feedback training" ↗

M. Barry Stermann & L. Friar, *Electroencephalography and Clinical Neurophysiology* 33(1) (1972)

The 1972 paper that showed seizure suppression is teachable. The founding empirical result of clinical neurofeedback.

"Über die strahlungslosen Eigenschwingungen einer leitenden Kugel, die von einer Luftschicht und einer Ionosphärenhülle umgeben ist" ↗

Winfried Otto Schumann, *Zeitschrift für Naturforschung* 7a (1952)

Schumann's prediction of the Earth-ionosphere resonance at ~7.83 Hz. The planet has a fundamental tone — measurable, repeatable, tuned to the same low band the brain inhabits. Read the Schumann resonance explainer →

"The Nervous System in the Context of Information Theory"

M. Zimmermann, in *Human Physiology* (Schmidt & Thews, eds.) (1989)

The textbook estimate that the nervous system handles roughly 10^7 bits per second of input while consciousness processes about 50 — the empirical source of the 40-bit paradox.

7. The Field hypothesis — consciousness as reception

Irreducible: Consciousness, Life, Computers, and Human Nature

Federico Faggin · [Amazon ↗](#)

The architecture under the trilogy. Faggin invented the microprocessor and spent twenty years concluding that consciousness cannot be reduced to computation. The three irreducible properties — knowing, choosing, feeling — are the chord.

The Idea of the World

Bernardo Kastrup · [Amazon ↗](#)

Analytical idealism, rigorously argued. Kastrup is referenced directly in *Anima's* "Field" chapter — universal mind as the most parsimonious explanation for the anomalous evidence.

The Emperor's New Mind & Shadows of the Mind

Roger Penrose · [Amazon ↗](#)

Orch-OR theory and the non-computable thesis. Penrose appears in *Anima's* "Instrument" chapter — the argument that consciousness cannot be a classical algorithm. The empirical revival of Orch-OR came in 2022, when Bandyopadhyay & Hameroff demonstrated quantum-coherent microtubule vibrations at biologically relevant temperatures. Read the Bandyopadhyay-Hameroff microtubule explainer → · Read the quantum-biology survey →

Being You: A New Science of Consciousness

Anil Seth · [Amazon ↗](#)

The most rigorous contemporary alternative — controlled hallucination and biological naturalism. *Limen's* longest sustained dialogue is with Seth's argument.

The User Illusion: Cutting Consciousness Down to Size

Tor Nørretranders · [Amazon ↗](#)

The 40-bit paradox — eleven million bits per second of input, forty bits of awareness. The compression is the architecture. Referenced explicitly in *Anima*.

Alien Information Theory: Psychedelic Drug Technologies and the Cosmic Game

Andrew Gallimore · [Amazon ↗](#)

An information-theoretic framing of DMT experience and the geometry of "hyperspace" reception — consciousness as bandwidth, the psychedelic state as a re-tuning of the antenna toward channels normally filtered out.

Irreducible Mind: Toward a Psychology for the 21st Century

Edward F. Kelly et al. · [Amazon ↗](#)

The University of Virginia collective's foundational compendium of the post-materialist case — 800 pages of anomalous evidence, marshalled rigorously. The reference work behind much of *Anima's* framing.

Beyond Physicalism: Toward Reconciliation of Science and Spirituality

Edward F. Kelly et al. · [Amazon ↗](#)

The sequel to *Irreducible Mind* — the case for non-physicalist consciousness made by working scientists rather than philosophers.

The Science Delusion: Freeing the Spirit of Enquiry

Rupert Sheldrake · [Amazon ↗](#)

Ten dogmas of materialism, examined and dismantled. *Limen's* methodological license to ask the questions the trilogy asks.

Determined: A Science of Life Without Free Will

Robert Sapolsky · [Amazon ↗](#)

The most thorough contemporary case for hard determinism — and the position *Limen's* freedom chapter is written against, fairly and at length.

The Elegant Universe: Superstrings, Hidden Dimensions, and the Quest for the Ultimate Theory

Brian Greene · [Amazon ↗](#)

The accessible map of string theory and extra dimensions — the cosmological frame within which *Limen's* nested-reality architecture lives.

Quantum Gravity

Carlo Rovelli · [Amazon ↗](#)

Loop quantum gravity and the relational interpretation. Space and time as derivative rather than fundamental — *Limen's* ontology in physics-textbook form.

The Quantum Theory of Fields (3 vols.)

Steven Weinberg · [Amazon ↗](#)

The Nobel laureate's canonical three-volume treatment. The textbook against which any field hypothesis of consciousness must measure its own seriousness.

Wholeness and the Implicate Order

David Bohm · [Amazon ↗](#)

Bohm's 1980 philosophical-physics synthesis — the implicate-order metaphysics that anticipated the contemporary field-cosmology programs by decades. The explicate spacetime world unfolds from a deeper, holographically-enfolded implicate order; matter and consciousness are two aspects of the same underlying totality (the *holomovement*). The philosophical-physics ancestor of D'Ariano-Faggin, Strømme, and Hoffman.

Brain and Perception: Holonomy and Structure in Figural Processing

Karl Pribram · [Amazon ↗](#)

Pribram's mature statement (Erlbaum, 1991) of the holonomic brain theory: the cortex as a Fourier transformer storing memory holographically rather than locally. The neuroscientific cognate of Bohm's implicate order — together, the Bohm-Pribram framework is the mid-twentieth-century precursor of the trilogy's receiver model.

Mindful Universe: Quantum Mechanics and the Participating Observer & Mind, Matter and Quantum Mechanics

Henry Stapp · [Amazon ↗](#)

The most carefully argued contemporary defense of the neo-Heisenbergian interpretation of quantum mechanics. Stapp studied under Pauli and Heisenberg; spent his career at LBNL. His central claim: orthodox quantum mechanics already requires conscious participation (von Neumann's Process 1 selection) as a structural feature, and the quantum Zeno effect supplies a precise mechanism for willed action on neural dynamics.

"The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs" ↗

Robin L. Carhart-Harris et al., *Frontiers in Human Neuroscience* 8 (2014)

The 2014 paper that organized contemporary psychedelic neuroscience around a single hypothesis: conscious states sit on an entropy axis, with the default-mode network's filtering corresponding to low-entropy operating ranges. Psychedelics, deep meditation, REM dreaming, and certain pathologies all relax the filter and reveal higher-entropy modes normally suppressed. The most direct contemporary neuroscientific ally of the trilogy's receiver model. The 2019 REBUS extension (with Karl Friston) supplies the predictive-processing mechanism.

8. The anomalous evidence

Children Who Remember Previous Lives & Twenty Cases Suggestive of Reincarnation

Ian Stevenson · [Amazon](#) ↗

Forty years of clinical-grade case documentation, mostly from the University of Virginia. The dataset under the Lucía Reyes thread in *Anima*.

Consciousness Beyond Life

Pim van Lommel · [Amazon](#) ↗

The cardiologist's prospective study of near-death experience in cardiac arrest survivors, published in *The Lancet*. The Mary Parker chapter rests on this dataset.

Threshold: Terminal Lucidity and the Border of Life and Death (2023)

Alexander Batthyány · [Amazon](#) ↗

Batthyány's book-length treatment of terminal lucidity cases — patients with destroyed cortical tissue regaining full coherence before death. The Mr. Martinez scene. Read about the research →

Cymatics: A Study of Wave Phenomena

Hans Jenny · [Amazon](#) ↗

Black sand on a white plate. The visual basis of the chord scene in *Numen* and the antenna model in *Limen*.

Chladni's plates · sound figures (1787) →

Ernst Chladni

The original empirical demonstration that frequency creates geometry. The conceptual ancestor of every cymatic image in *Limen*. See the historical plates →

The Fourth Phase of Water: Beyond Solid, Liquid, and Vapor

Gerald H. Pollack · [Amazon](#) ↗

Exclusion-zone water as the substrate where biology meets electromagnetism. The hydrologic ground of *Limen*'s antenna model.

Islands of Genius: The Bountiful Mind of the Autistic, Acquired, and Sudden Savant

Darold A. Treffert · [Amazon ↗](#)

Book-length treatment of savant abilities — including sudden savant syndrome, the rarest and most theory-disrupting variant. A reference for *Numen*'s capacity-emergence scenes.

9. Frequency, geometry, the golden ratio

The Power of Limits: Proportional Harmonies in Nature, Art, and Architecture

György Doczi · [Amazon ↗](#)

Phi proportion across biology, architecture, and sound. The book that taught the author to look for the chord. Read the ϕ explainer →

On the Sensations of Tone as a Physiological Basis for the Theory of Music

Hermann von Helmholtz · [Amazon ↗](#)

The 19th-century foundational treatise (1863; trans. A.J. Ellis) on the physics and physiology of musical hearing — overtones, dissonance, and the body's tuned response to frequency. The acoustic ground beneath the chord's psychology.

How Music Works

John Powell · [Amazon ↗](#)

A working tuner's-eye view of why equal temperament is a compromise — and why a chord tuned to exact ratios sounds the way it does.

This Is Your Brain on Music

Daniel Levitin · [Amazon ↗](#)

The neuroscience of resonance, written for clinicians who already suspected the body was doing more than processing.

A New Kind of Science

Stephen Wolfram · [Amazon ↗](#)

Computational irreducibility — simple rules producing irreducible complexity. The mathematical mechanism behind *Limen*'s nested-reality architecture.

The Golden Ratio: The Story of Phi, the World's Most Astonishing Number ↗

Mario Livio (*Broadway Books*, 2002)

The canonical accessible treatment of ϕ . Livio is an astrophysicist who set out to write a ϕ -celebration book and ended up writing the most careful debunking of ϕ -mythology in print — without diminishing the parts of the story that are real. The history (Euclid, Fibonacci, Pacioli, Kepler), the mathematics (continued fractions, regular pentagon, Fibonacci convergence), the biology (phyllotaxis, log-spirals), and the much smaller-than-folklore-claims footprint of ϕ in art and architecture. The single book the trilogy assumes the reader has touched.

Phyllotaxis: A Systemic Study in Plant Morphogenesis

Roger V. Jean · [Amazon ↗](#)

The mathematical anatomy of why plants count in Fibonacci. The botany under the trilogy's geometry chapters.

The Algorithmic Beauty of Plants

Przemyslaw Prusinkiewicz & Aristid Lindenmayer · [Amazon ↗](#)

L-systems — recursive grammars that grow ferns, trees, and inflorescences. *Limen's* argument that biology is computation reading geometry.

***De Divina Proportione* (illustrated by Leonardo da Vinci, 1509)**

Luca Pacioli · [Amazon ↗](#)

The Renaissance treatise that named the divine proportion. Where Leonardo's polyhedra meet the trilogy's claim that beauty has a structure.

The Science of Sound

Thomas D. Rossing · [Amazon ↗](#)

The standard acoustics textbook. The technical ground beneath every chord scene in the trilogy.

10. Composers — frequency, the harmonic series, and the dissolution of the 12-tone grid

The twentieth- and twenty-first-century composers who took seriously what the trilogy's frequency claim makes audible: frequency as continuous architecture rather than as a coordinate on a grid. Each entry has a longer treatment on the Watch & Listen page.

György Ligeti · *Atmosphères* (1961) & *Lux Aeterna* (1966)

[Watch & Listen ↗](#)

The micropolyphonic cluster works Kubrick chose for the monolith and stargate scenes in *2001: A Space Odyssey*. Dozens of independent slow voices layered into a continuous frequency field with no melody, no harmony, only the sonic mass itself. The 12-tone grid dissolves into pure field — the sonic correlate of *Numen's* chord that refuses to resolve.

Gérard Grisey · *Partiels* (1975)

[Watch & Listen ↗](#)

The founding work of *spectral music*. Grisey takes a single low trombone E and rebuilds it orchestrally as its own overtone series. The sound's internal architecture made audible. The school (Grisey, Murail, Saariaho, Rădulescu) takes seriously what the trilogy says: frequency is structure, not coordinate; a "note" is a stack of overtones the body hears as a whole.

Tristan Murail · *Gondwana* (1980)

[Watch & Listen ↗](#)

The second canonical spectral work. Murail derives the entire orchestral piece from the analyzed spectrum of a bell tone. A bell heard from inside, expanded over twenty minutes.

Kaija Saariaho · *Lichtbogen* (1986)

[Watch & Listen ↗](#)

The most lyrical of the spectral works. Cello harmonic plus live electronics extending the natural overtones; the title comes from the Northern Lights Saariaho saw on a flight back to Finland. Where Grisey shows the internal architecture, Saariaho lets you walk through it. The cleanest entry point for a listener new to the spectral school.

Horăţiu Rădulescu · *String Quartet No. 4 «Infinite to be cannot be infinite, infinite anti-be could be infinite» (1976-87)*

Watch & Listen ↗

128 strings, scordatura-tuned to the harmonic series, four players each alternating between two instruments. Rădulescu called the approach *sound plasma*. The closest pre-trilogy musical realization of *the body inside the field*: not metaphorical, sonically literal.

Iannis Xenakis · *Metastaseis (1953-54)*

Watch & Listen ↗

Xenakis was an architect (Le Corbusier, La Tourette, Philips Pavilion) before he was a composer. *Metastaseis* translates the Modulor's golden-ratio proportions into orchestral pitch space — 61 strings on independent glissandi tracing a hyperboloid surface. The ϕ -spiral as orchestral score.

Giacinto Scelsi · *Quattro Pezzi per Orchestra (su una nota sola) (1959)*

Watch & Listen ↗

Four orchestral pieces, each built entirely from a single note. The trilogy's "one note has internal architecture" claim, enunciated more radically than the later spectral school would enunciate it.

Karlheinz Stockhausen · *Stimmung (1968)*

Watch & Listen ↗

75 minutes of six voices sustaining a single B $\flat$ 9 overtone-chord, exploring its interior through vowel formants. No new chord ever introduced; the piece *is* the inside of one harmonic structure, sustained long enough for the listener to walk around inside it. Explicitly a meditation: the singers chant deity names from many traditions across the work.

Glenn Branca · *Symphony No. 3 «Gloria — Music for the First 127 Intervals of the Harmonic Series» (1983)*

Watch & Listen ↗

Guitar orchestra performing literally the first 127 intervals of the natural harmonic series, on custom-built instruments tuned to those overtones. What standard notation calls "out of tune" Branca treats as the actual structure of sound. The rock/orchestral pole of the spectral school.

La Monte Young · *The Well-Tuned Piano (1964-)*

Watch & Listen ↗

Young's lifelong piece. Grand piano retuned to 7-limit just intonation; performed in single sittings of five to six hours. Long enough that the notes stop being notes and start being the room they make. The head of the contemporary drone-and-just-intonation lineage Catherine Lamb and Catherino now extend.

Catherine Lamb · *prisma interius series and other just-intonation works (2010s-)*

Watch & Listen ↗

The youngest voice in the just-intonation lineage. Long-form, intimate, built from precisely-tuned harmonic intervals — sound treated with the patience of botany. Often performed by Konzert Minimal in Berlin. The contemporary continuation of Young's framework.

Dolores Catherino · polychromatic music in 31, 72, and 106 tones per octave (2010s-)

Watch & Listen ↗

The contemporary listening room closest to the trilogy's ϕ -tuned C. 72-TET contains the ϕ ratio within 0.2 cents (step 50 = 833.3¢ vs. true ϕ at 833.1¢); 106-TET dissolves discrete pitch entirely into continuous frequency. The cleanest recording of what *Limen*'s chord actually sounds like at the literal frequencies.

11. Quantum biology — the body as quantum-coherent system

The four-case empirical foundation for the claim that biology exploits quantum-mechanical processes at body temperature. See the quantum-biology survey explainer →

"Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems" ↗

Gregory S. Engel et al., Nature 446 (2007)

The 2007 paper that opened the field. Two-dimensional electronic spectroscopy of the Fenna-Matthews-Olson (FMO) photosynthetic complex demonstrating quantum-coherent energy transfer at femtosecond timescales. Photosynthesis as a quantum-mechanical search algorithm. The first decisive result in quantum biology. Read the quantum-biology survey →

"The Radical-Pair Mechanism of Magnetoreception" ↗

P. J. Hore & Henrik Mouritsen, Annual Review of Biophysics 45 (2016)

The contemporary review of the radical-pair mechanism by which avian and other animal magnetoreception works. Quantum-entangled electron spins in cryptochrome molecules in the retina, evolving under Earth's magnetic field, surviving long enough (microseconds) to function as a biological compass. The cleanest in vivo demonstration of biologically functional quantum entanglement at room temperature. Read the quantum-biology survey →

Microtubule quantum coherence at biologically relevant temperatures

Anirban Bandyopadhyay, Stuart Hameroff, et al. (NIMS, c. 2014–2024)

A decade-plus experimental program at NIMS (Tsukuba) demonstrating discrete coherent resonance peaks in single microtubules and microtubule networks spanning kHz through THz, surviving thermal averaging at body temperature. The 2022 results in particular revived Orch-OR as a serious empirical hypothesis: the warm-wet-noisy objection no longer holds when the substrate turns out to be a self-protecting resonator. Combined with the Pollack EZ-water work, the microtubule-water composite is the strongest current candidate for the cellular-scale receiver.

Life on the Edge: The Coming of Age of Quantum Biology

Jim Al-Khalili & Johnjoe McFadden · Amazon ↗

The accessible non-technical overview of the field. Al-Khalili (theoretical physicist) and McFadden (molecular biologist) walk through photosynthesis, magnetoreception, olfaction, and microtubule coherence for a general audience. The single-book introduction. See the quantum-biology survey →

12. The mystics — phenomenological convergence

Interior Castle

Teresa of Ávila · [Amazon ↗](#)

Seven dwellings of the soul. The structural inspiration for the threefold movements of *Numen*, and a primary text in *Limen*'s mystics chapter. For Teresa read phenomenologically through the *Libro de la Vida*, see the Teresa explainer →

Dark Night of the Soul & Spiritual Canticle

Juan de la Cruz · [Amazon ↗](#)

The receiver being recalibrated. José's nightstand book in *Anima*.

The Essential Rumi (trans. Coleman Barks)

Jalāl al-Dīn Rumi · [Amazon ↗](#)

The reed flute — the antenna model expressed as poetry eight centuries early. *You are not a drop in the ocean, but the ocean in a drop.*

Sermons & Treatises

Meister Eckhart · [Amazon ↗](#)

The *Gotttheit* — Layer 0 without reception. The contemplative ground beneath any consciousness framework that takes the field seriously.

The Principal Upanishads

trans. S. Radhakrishnan · [Amazon ↗](#)

Sat-chit-ananda — being, consciousness, bliss — mapped onto Faggin's three properties and the augmented chord in *Limen*.

Tantrāloka (selections)

Abhinavagupta · [Amazon ↗](#)

The Kashmir Shaivite root of Luz Paz's contemplative framework in *Fragile Light* — cited in her university admissions essay. Abhinavagupta's massive synthesis (thirty-seven chapters, several thousand verses) of the Trika, Pratyabhijñā, Krama, and Spanda schools into a single coherent philosophy. The technical home of the five-shakti analysis (Cit, Ānanda, Icchā, Jñāna, Kriyā) whose middle three are a word-for-word cognate of Faggin's knowing-choosing-feeling triad. Read the Kashmir Shaivism explainer — five shaktis, Pratyabhijñā, spanda, and the trilogy's contemplative architecture →

The Mystical Thought of Meister Eckhart

Bernard McGinn · [Amazon ↗](#)

The standard scholarly study of Eckhart's mystical theology. McGinn is the historian to read alongside Eckhart's own sermons.

The Darkness of God: Negativity in Christian Mysticism

Denys Turner · [Amazon ↗](#)

The apophatic tradition examined philosophically — the *via negativa* as logic, not evasion. The ground beneath *Limen*'s claim that the field is best described by what it isn't.

The Sufi Path of Love: The Spiritual Teachings of Rumi

William C. Chittick · [Amazon ↗](#)

The scholarly companion to Coleman Barks's freer renderings. The Akbarian frame beneath Rumi's poetry.

Mystical Dimensions of Islam

Annemarie Schimmel · [Amazon ↗](#)

Schimmel's standard survey of Sufism — the historical and theological breadth Rumi appears within.

The Upanishads (trans. Eknath Easwaran)

Eknath Easwaran · [Amazon ↗](#)

The accessible spiritual-practitioner translation. The pairing to Radhakrishnan's scholarly edition — Easwaran is what one reads in the morning.

The Early Upanishads: Annotated Text and Translation

Patrick Olivelle · [Amazon ↗](#)

The Oxford scholarly edition — the critical-text foundation for any serious engagement with the Upanishads in English.

Vimalakīrti Sūtra (Burton Watson trans.)

anonymous Mahayana, c. 100 CE · [Amazon ↗](#)

The dog-eared copy on Luz Paz's shelves in *Fragile Light* — the Mahayana sutra whose lay protagonist out-argues the bodhisattvas, and whose silence at the question of nonduality answers it more fully than speech could.

Śiva Sūtras: The Yoga of Supreme Identity (Jaideva Singh trans.)

ascribed to Vasugupta, c. 9th century CE · [Amazon ↗](#)

The root text of Kashmir Shaivism — the aphorisms Abhinavagupta later expounded. Luz read the *Śiva Sūtras* at fifteen, alone in Galicia. Read the Kashmir Shaivism explainer →

13. Cognition, attention, and the divided brain

The Master and His Emissary & The Matter With Things

Iain McGilchrist · [Amazon ↗](#)

The hemispheric architecture of attention. *Limen's* argument that the production model is what the modeling hemisphere insists on when it has stopped listening.

Awakening from the Meaning Crisis (lecture series)

John Vervaeke · [Amazon ↗](#)

Relevance realization — why biology is non-negotiable for genuine rationality. The reason post-human hybrids in *Numen* need biological substrate not for power but for stakes.

The Spell of the Sensuous

David Abram · [Amazon ↗](#)

Phenomenology of perception in living, breathing prose. Underlies the way *Anima* describes the field.

Consciousness Explained

Daniel Dennett · [Amazon ↗](#)

The position any field hypothesis has to answer. Dennett is fair company even when one disagrees, and *Limen* engages him directly on the linguistic-self-narration argument.

Thought and Language

Lev Vygotsky · [Amazon ↗](#)

Inner speech as the mechanism by which the antenna recursively modifies its own architecture. The root text of *Limen*'s "language as antenna tuning" chapter.

Language and Mind

Noam Chomsky · [Amazon ↗](#)

Recursion as the Faculty of Language in the Narrow sense. Hauser, Chomsky, and Fitch on what is uniquely human.

Supersizing the Mind

Andy Clark · [Amazon ↗](#)

Words as cognitive artifacts — the extended mind. Clark's argument that the antenna doesn't stop at the skin.

How Language Shapes the Way We Think (TED) & "Linguistic Relativity"

Lera Boroditsky · [Amazon ↗](#)

The Kuuk Thaayorre speakers who maintain constant cardinal-direction orientation — evidence that language physically rewires what the antenna can hear.

Becoming Human: A Theory of Ontogeny

Michael Tomasello · [Amazon ↗](#)

Cooperative cognition as shared tuning across receivers — the social mechanism of human consciousness.

Descartes' Error: Emotion, Reason, and the Human Brain

Antonio Damasio · [Amazon ↗](#)

Damasio's argument that emotion is not a contaminant of reason but its scaffolding. The neurology of feeling as cognition.

An Odd Kind of Fame: Stories of Phineas Gage

Malcolm Macmillan · [Amazon ↗](#)

The most rigorous historical reconstruction of the Gage case — what we actually know vs. what neurology has projected onto him.

Permanent Present Tense: The Unforgettable Life of the Amnesic Patient, H.M.

Suzanne Corkin · [Amazon ↗](#)

Five decades with H.M. — the patient whose loss of memory taught us how memory is made.

The Social Brain: Discovering the Networks of the Mind

Michael S. Gazzaniga · [Amazon ↗](#)

The interpreter module — the left hemisphere as confabulator. The technical ancestor of *Limen*'s account of narrative self-construction.

The Voices Within: The History and Science of How We Talk to Ourselves

Charles Fernyhough · [Amazon ↗](#)

Inner speech, studied empirically. The contemporary companion to Vygotsky — the clinical ground beneath the trilogy's language chapters.

14. Medicine, attention, and the clinical encounter

How Doctors Think

Jerome Groopman · [Amazon ↗](#)

The quality of attention a physician owes a patient. The book under *Anima*'s clinical scenes.

Being Mortal

Atul Gawande · [Amazon ↗](#)

What the body asks for at the end. What the system asks for instead. The ethical ground of the Mr. Martinez and Ray Montoya chapters.

The Body Keeps the Score

Bessel van der Kolk · [Amazon ↗](#)

The body as the most reliable instrument we have for telling the truth about what arrived.

Musicophilia: Tales of Music and the Brain

Oliver Sacks · [Amazon ↗](#)

Sacks at his best — the clinical phenomenology of music in the brain, told one patient at a time.

An Anthropologist on Mars: Seven Paradoxical Tales

Oliver Sacks · [Amazon ↗](#)

Seven case histories of unusual minds — and the model of clinical writing the trilogy aspires to.

The Polyvagal Theory: Neurophysiological Foundations of Emotions, Attachment, Communication, and Self-Regulation

Stephen W. Porges · [Amazon ↗](#)

The vagal architecture of safety. The neurobiology beneath the clinical encounter — what the body does when it feels heard.

"In vitro neurons learn and exhibit sentience when embodied in a simulated game-world" (DishBrain) ↗

Kagan et al., *Neuron* 110(23) (2022)

Human neurons learning to play Pong on a multielectrode array. The published evidence that the hybrid arc of *Numen* is engineering, not speculation.

Brainware, FinalSpark, Cortical Labs CL1 — three approaches to biocomputing (2023-2025)

[Cortical Labs CL1 product page ↗](#) · [Kalil overview ↗](#)

Three parallel attempts to build computation out of living human neurons. **Brainware** (Indiana University, 2023) is the academic frontier: 3D human brain organoids interfaced with microelectrode arrays, demonstrating pattern recognition and signal processing. **FinalSpark** (Swiss startup) is the cloud play: neuron cultures accessed remotely, framed against the energy gap between brains (~20 W for 86 billion neurons) and megawatt-scale GPU clusters. **Cortical Labs CL1** is the productized version, launched at Mobile World Congress Barcelona in 2025: ~800,000 human neurons on a silicon microelectrode array with on-chip life support (pumps, gas, temperature), running a "Biological Intelligence OS" that lets researchers deploy code to the neurons. Around \$35K, six-month neuron lifespan, animal-free testbed for drug discovery, AI research, robotics. The empirical floor under *Numen*'s hybrid arc with Sable. Read the consolidated wetware explainer →

Neuralink CONVOY Study & adaptive deep brain stimulation (aDBS)

[Neuralink CONVOY launch ↗](#) · 2023-2026

The two ends of the closed-loop neurotechnology spectrum, both in clinical use by 2025. **Neuralink's CONVOY** ("Control of Assistive Devices Via Brain-Computer Interface Technology") is a prospective early-feasibility trial in PRIME-implanted participants with quadriplegia, decoding motor-cortex activity into control of robotic arms and computers — read-out only, no stimulation back into the brain. Public demonstrations include a participant feeding himself with a brain-controlled arm and achieving mouse-equivalent cursor bit-rates. **Medtronic's BrainSense Adaptive DBS** is the first commercial closed-loop neurostimulator: sensing local field potentials in the subthalamic nucleus or GPi and adjusting stimulation amplitude in real time to track the patient's brain state. FDA-approved in the mid-2020s, with more than 2,000 Parkinson's recipients by 2026. CONVOY *reads* the brain; aDBS *reads and writes* to deep nuclei. Together they bracket the technical horizon Sable inhabits in *Numen*. Read the consolidated wetware + BCI explainer →

15. Ethics, freedom, and the love-freedom connection

Summa Theologiae (selections)

Thomas Aquinas · [Amazon ↗](#)

Amor est velle alicui bonum — to love is to will the good of the other for their own sake. The epigraph of *Anima*, and the working definition of love throughout the trilogy.

Political Ponerology

Andrzej Łobaczewski · [Amazon ↗](#)

Hierarchical systems select for pathological personalities the way a wound attracts infection. The argumentative spine of Luz's exchanges with Jordi in *Fragile Light*.

The Kingdom of God Is Within You

Leo Tolstoy · [Amazon ↗](#)

The withdrawal of obedience as the mechanism of every nonviolent transformation. Bodhi cites it on the patio.

Live Not by Lies

Aleksandr Solzhenitsyn · [Amazon ↗](#)

The Soviet system sustained not by tanks but by daily compliance. The conversation under the stars in *Fragile Light*.

The Subversion of Christianity & Anarchy and Christianity

Jacques Ellul · [Amazon ↗](#)

Christian anarchism. The third leg of Luz and Bodhi's voluntarist conversation.

Reasons and Persons

Derek Parfit · [Amazon ↗](#)

Personal identity, simulation, the ethics of bringing beings into existence. The undertext of *Numen*'s final movement.

No Treason: The Constitution of No Authority (1870)

Lysander Spooner · [Amazon ↗](#)

The 19th-century natural-law argument that no person was ever bound by a constitution they did not sign. The leftmost book on Luz Paz's voluntarist shelf in *Fragile Light*.

Our Enemy, the State (1935)

Albert Jay Nock · [Amazon ↗](#)

Nock's distinction between social power and state power, and the doctrine of the Remnant. Cited by name in Luz's exchanges with Jordi.

The Market for Liberty (1970)

Morris & Linda Tannehill · [Amazon ↗](#)

The market-anarchist case — voluntary institutions modeling what the state claims only coercion can produce. The Tannehills sit between Nock and the *Vimalakīrti Sūtra* on Luz's shelves.

The Most Dangerous Superstition

Larken Rose · [Amazon ↗](#)

Rose's argument that the belief in authority is the foundational error of political life. The contemporary voice in Luz's voluntarist library — and one of the texts Bodhi was trained on.

16. Theology — Ground of Being

Systematic Theology, vol. I (esp. "Ground of Being")

Paul Tillich · [Amazon ↗](#)

The closest philosophical analogue to the consciousness field, named with care. *Limen*'s theology section opens here.

I and Thou

Martin Buber · [Amazon ↗](#)

The I-Thou distinction. Theism requires personal encounter; the framework permits but does not require it. The question is held open, honestly.

Metaphysics, Book IV (c. 350 BCE)

Aristotle · [Amazon ↗](#)

The principle of non-contradiction and the inquiry into being-as-being. The philosophical ground on which any consciousness ontology has to stand.

Republic (Book VI) & Symposium (210a-212b)

Plato · [Amazon ↗](#)

The divided line and the ladder of beauty — Plato's two great staircases up from appearance to form. The pre-Socratic ancestor of every Ground-of-Being argument.

Enneads, I.6 ("On Beauty") and V.1 ("On Emanation from the One")

Plotinus · [Amazon ↗](#)

Plotinus on the One as source. The Neoplatonic frame Christian and Sufi mysticism would inherit — and the contemplative root beneath Eckhart's *Gotttheit*.